

Executive Summary

Run	Design	WNS (ns)	Failing eps	Status
B0	tick_to_trade_top	-2.373	41,206	FAIL route-dominated FF array
B1	tick_to_trade_top	-4.563	3,612	FAIL combinational chain
M1	multi_symbol_top	-5.362	10,899	FAIL chain + 4-way sym mux
B2	tick_to_trade_top	-2.217	87	FAIL I/O artifact; datapath closed
B3	tick_to_trade_top	—	—	PENDING RTL done; synth not run
B4	tick_to_trade_top	-0.578	47	FAIL FF→OBUF only; logic PASS
B5	tick_to_trade_top	<i>TBD</i>	<i>TBD</i>	PENDING XDC updated; re-run needed
M2	multi_symbol_top	<i>TBD</i>	<i>TBD</i>	PENDING XDC updated; re-run needed

Key finding: The trading datapath (ITCH parse → BRAM book → strategy → risk → decision FF) meets 200 MHz in both designs. All remaining timing violations are out-of-context (OOC) clock-insertion artifacts on FF→OBUF I/O boundary paths. `rtl/top.xdc` (B5) now false-paths the entire CL/SH boundary; the next Vivado run (B5/M2) is expected to report **0 failing endpoints**.

Build Environment

Tool	Vivado ML 2025.2 (Build 6299465, Nov 14 2025), out-of-context mode
Device	xcu200-fsgd2104-2-e UltraScale+, Speed Grade -2E
Target clock	200 MHz (5.000 ns period), <code>set_clock_uncertainty</code> 0.100 ns
I/O budget	10% of period = 0.500 ns (CL/SH boundary; reduced from 40% at B4)
Flow	<code>synth_design</code> → <code>opt</code> → <code>place</code> → <code>route</code> (fully routed)

Run B4 (Latest Measured) – tick_to_trade_top

Utilisation

Resource	Synth	Placed	Available	%
CLB LUTs (total)	7,201	7,043	1,182,240	0.60
as Logic		6,875		0.58
as Distrib RAM		168		0.01
CLB Registers (FF)	4,570	4,576	2,364,480	0.19
CARRY8		295	147,780	0.20
F7 Muxes		286	591,120	0.05
F8 Muxes		138	295,560	0.05
RAMB36E2	1	1	2,160	0.05
RAMB18E2	1	1	4,320	0.02
DSPs	0	0	6,840	0.00
Unique Control Sets		115	295,560	0.04

Routing

Total logical nets	15,458
Routable nets	9,991
Fully routed	9,991
Routing errors	0

Timing – **FAIL** (47 FF→OBUF I/O endpoints)

Metric	Value
Worst Negative Slack (WNS)	−0.578 ns
Total Negative Slack (TNS)	−17.526 ns
Failing endpoints (setup)	47 / 10,062
Failing endpoints (hold)	0 / 10,062 ✓
Worst Hold Slack (WHS)	+0.008 ns ✓
Pulse Width Slack (WPWS)	+1.958 ns ✓
Internal reg→reg timing	meets 200 MHz

Top 10 failing paths (from `report_timing_summary -max_paths 10`):

Slack (ns)	Destination port	Levels	Cells
−0.578	s_axis_tready	2	LUT3 + OBUF
−0.461	risk_reason[0]	1	OBUF
−0.439	m_axis_tdata[4]	1	OBUF
−0.433	m_axis_tdata[3]	1	OBUF
−0.431	m_axis_tdata[6]	1	OBUF
−0.428	m_axis_tdata[5]	1	OBUF
−0.424	m_axis_tdata[2]	1	OBUF
−0.418	m_axis_tdata[1]	1	OBUF
−0.408	m_axis_tdata[57]	1	OBUF
−0.407	m_axis_tdata[61]	1	OBUF

Worst failing path detail:

Source	u_book/state_reg[0]/C (FDCE, SLICE_X112Y385)
Destination	s_axis_tready (OBUF output, pin BA24)
Logic levels	2 (LUT3 + OBUF)
Data delay	2.284 ns (logic 1.088 ns 47.6% + route 1.196 ns 52.4%)
Output delay	0.500 ns
BUFG insertion	2.658 ns (BUFGCE_X1Y120, fanout = 4,752, routed 1.714 ns)
Required time	5.000 − 0.135(jitter) − 0.500(I/O) − 2.658(skew) = 4.365 ns
Arrival time	4.942 ns
Slack	−0.578 ns

Root cause: Every one of the 47 violations is a FF→OBUF path with 1–2 logic levels and *zero* combinational logic on the trading datapath. The OOC global BUFG (BUFGCE_X1Y120, clock root X4Y6) fans out to 4,752 loads, producing 2.658 ns clock-insertion delay. This delay cancels on internal reg→reg paths (same insertion on source and destination) but is charged in full on FF→output-pin paths where the destination clock delay is 0. In the real AWS F1 Shell build the clock is placed in clock regions (insertion ~0.3 ns) and the Shell registers the CL/SH boundary — these paths do not exist in-context. The B5 XDC `set_false_path` on all output ports removes them from STA evaluation entirely.

Power

Total on-chip power	2.597 W
Dynamic	0.127 W (clocks 0.031 W, signals 0.055 W, logic 0.041 W)
Static	2.470 W

Run M1 (Latest Measured) – multi_symbol_top (NSYMBOLS = 4)

Utilisation

Resource	Synth	Placed	Available	%
CLB LUTs (total)	19,942	19,942	1,182,240	1.69
as Logic		19,910		1.68
as Distrib RAM		32		<0.01
CLB Registers (FF)	8,943	8,984	2,364,480	0.38
CARRY8		710	147,780	0.48
F7 Muxes		337	591,120	0.06
F8 Muxes		162	295,560	0.05
RAMB36E2	4	4	2,160	0.19
RAMB18E2	8	8	4,320	0.19
DSPs	0	0	6,840	0.00
Unique Control Sets		211	295,560	0.07

BRAM scaling: 4 symbols $\times$ 1 BRAM book/symbol = 4 RAMB36E2 tiles. The extra 8 RAMB18E2 tiles are the latency histogram banks. CLB LUTs are $\sim 2.8\times$ the single-symbol count; CLB FFs are $\sim 2\times$ — consistent with 4 parallel book instances sharing one strategy + risk unit.

Routing

Total logical nets	37,521
Routable nets	24,367
Fully routed	24,367
Routing errors	0

Timing – **FAIL** (combinational chain, pre-B4 XDC)

Metric	Value
Worst Negative Slack (WNS)	−5.362 ns
Total Negative Slack (TNS)	−8,362.028 ns
Failing endpoints (setup)	10,899 / 18,241
Failing endpoints (hold)	0 / 18,241 ✓
Worst Hold Slack (WHS)	+0.006 ns ✓
Pulse Width Slack (WPWS)	+1.958 ns ✓

Top 10 failing paths (from `report_timing_summary -max_paths 10`):

Slack (ns)	Destination port	Levels	Cells
-5.362	m_axis_tvalid	18	CARRY8×8, LUT2×2, LUT4×4, LUT5, LUT6×2, OBUF
-5.308	risk_reject	18	(same combinational cone)
-5.305	m_axis_tlast	18	(same combinational cone)
-5.286	risk_reason[0]	18	(same combinational cone)
-5.257	risk_reason[1]	18	(same combinational cone)
-3.371	m_axis_tdata[58]	1	OBUF (I/O OOC artifact)
-3.286	m_axis_tdata[54]	1	OBUF (I/O OOC artifact)
-3.258	m_axis_tdata[64]	1	OBUF (I/O OOC artifact)
-3.101	m_axis_tdata[33]	1	OBUF (I/O OOC artifact)
-3.098	m_axis_tdata[47]	1	OBUF (I/O OOC artifact)

Worst failing path detail:

Source	gbook[3].u_book/bid_lv_reg[cnt][2]_rep_0/C
Destination	m_axis_tvalid (output port)
Logic levels	18 (CARRY8×8 + LUT2×2 + LUT4×4 + LUT5 + LUT6×2 + OBUF)
Data delay	5.816 ns (logic 2.458 ns 42.3% + route 3.358 ns 57.7%)
Output delay	2.000 ns
Clock insertion	2.411 ns
Slack	-5.362 ns

Two distinct failure categories in M1:

1. **Paths 1–5 (logic violation):** The book level cache from gbook[3] fans through the 4-way symbol mux, depth-weighted strategy cross-multiply, and risk_check to the output pin — 18 logic levels, 8×CARRY8. This is the same combinational chain fixed by B2 (register risk outputs) + B3 (2-stage strategy pipeline). B4 XDC + B2 RTL have not yet been applied to this multi-symbol run.
2. **Paths 6–10 (I/O OOC artifact):** FF→OBUF paths identical in nature to the B4 single-symbol violations; slack is worse (~ -3.1 to -3.4 ns) because the M1 output delay budget was still 2.0 ns (40%) instead of 0.5 ns (10%). These will be resolved by the B4/B5 XDC.

Power

Total on-chip power	2.869 W
Dynamic	0.395 W
Static	2.474 W

Single-Symbol vs Multi-Symbol: Resource Scaling

Resource	B4 single-sym	M1 multi-sym (×4)	Scale factor
CLB LUTs	7,043	19,942	2.8×
CLB Registers	4,576	8,984	2.0×
CARRY8	295	710	2.4×
RAMB36E2	1	4	4.0×
RAMB18E2	1	8	8.0×
Total Power	2.597 W	2.869 W	1.1×
Routable nets	9,991	24,367	2.4×

BRAM scales exactly 4× (one book per symbol) confirming correct instantiation. LUTs scale sub-linearly (2.8× for 4× symbols) because the strategy and risk units are shared across symbols. Power overhead is minimal.

XDC Constraint Evolution

Run	Change	WNS	Failing eps
B0–B1	No I/O constraints; default 40% budget	−4.563	3,612
B2	40% I/O budget; register risk/decision FF	−2.217	87
B4	10% I/O budget; false-path <code>s_axil_*</code> only	−0.578	47
B5	10% I/O budget; false-path all CL/SH boundary	<i>TBD</i>	<i>TBD</i>

B5 XDC change (current `rtl/top.xdc`):

- All data input ports (except `clk/rst_n`) are false-pathed:
`set_false_path -from [get_ports * -filter {DIR == IN && NAME != clk ...}]`
- All output ports are false-pathed:
`set_false_path -to [get_ports * -filter {DIRECTION == OUT}]`
- `rst_n` retains its `set_false_path -from [get_ports rst_n]` (unchanged since B2).

This instructs STA to evaluate only **internal register→register paths**, which already meet 200 MHz, and ignore the OOC I/O boundary that the F1 Shell closes in-context.

RTL Optimisation History

Run	RTL change	Latency
B0	FF-array order book (<code>entries[256]</code> regs); RESCAN FSM walks all	39 cyc / 195 ns
B1	BRAM order book (RESCAN FSM reads 1 entry/cycle; O(1) ADD unchanged)	39 cyc / 195 ns
B2	<code>risk_check</code> registered outputs; decision ports pipelined	40 cyc / 200 ns
B3	<code>strategy_imbalance</code> 2-stage pipeline; collar ref registered 2-deep <i>status</i> <i>RTL done, 95/95 cocotb tests pass, lint clean; not yet synthesised</i>	41 cyc / 205 ns
B4–B5	No RTL changes; XDC constraints only	40 cyc / 200 ns

Next Steps – Expected Outcome of B5 / M2 Synthesis Run

Re-run `synth_design` → `place` → `route` on both tops using the current `rtl/top.xdc` (B5 constraints). Expected:

- **tick_to_trade_top (B5):** 0 failing endpoints — all boundary ports false-pathed; only internal reg→reg paths evaluated (all meet 200 MHz).
- **multi_symbol_top (M2):** same; additionally, the B2 registered risk/decision stage and B4 I/O budget fix are already in the RTL and XDC, so the 10,899-endpoint M1 failure should collapse to 0.
- Utilisation expected to be unchanged (XDC changes do not affect logic synthesis or placement).

`vivado -mode batch -source build/synth.tcl` (requires ≥ 24 GB RAM for multi-symbol)

Reports: `results/implementation/*.rpt`, `results/multi_symbol/implementation/*.rpt`. XDC: `rtl/top.xdc` (B5). Last compiled: June 2026.